Tasbid Al Rahman

AI Systems Developer · ML Researcher

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Upwork Profile
Dhaka, Bangladesh

TECHNICAL SKILLS

Languages

Python, TypeScript, JavaScript, PHP, HTML/CSS

AI / ML

PyTorch, scikit-learn, Federated Learning, Quantum ML, PennyLane, Qiskit

Web / Backend

Next.js 14, Node.js, Express.js, REST APIs

Databases

PostgreSQL, MySQL, Prisma ORM

Automation / Tools

n8n, Git, Vercel, Cloudinary, Supabase

Data Science

NumPy, Pandas, Matplotlib, SciPy

EDUCATION

North South University

BSc in Computer Science and Engineering
2022 to Present, 4th Year

RESEARCH

Hybrid Classical-Quantum Split Federated Learning for IoMT Intrusion Detection.
Multi-university collaboration: NSU, BUET, KUET.
Target: Q1 Journal Publication

STRENGTHS

- Research and System Design
- Full-Stack Development
- AI Pipeline Engineering
- Academic Writing
- Cross-team Collaboration

AVAILABLE FOR HIRE

Open to freelance and contractual projects in AI/ML, automation, and web development. Active on Upwork and Fiverr.

PROFESSIONAL SUMMARY

Final-year CSE student at North South University, working on AI systems, machine learning, and full-stack development. I am currently running a multi-university research project on Hybrid Classical-Quantum Split Federated Learning for IoMT intrusion detection, targeting a Q1 journal. I have built healthcare platforms, e-commerce systems, and automation pipelines from the ground up, and I am open to freelance and contractual work.

PROJECTS

Hybrid Classical-Quantum Split Federated Learning

Research Project | NSU x BUET x KUET | In Progress, Q1 Journal Target

Python PyTorch Federated Learning Quantum ML (PennyLane, Qiskit) scikit-learn

- Building a privacy-preserving security framework for medical IoT devices, combining split learning, federated aggregation, and a quantum kernel layer for intrusion detection.
- Baseline Conv1D models and preprocessing done across four datasets: WESAD, PAMAP2, MIMIC-III, and TON-IoT. Currently implementing the split learning pipeline.
- Clients share only activation vectors with the server, keeping raw patient data on-device at all times.

Empathetic Full-Duplex Speech LLM (EFSM)

Deep Learning Research (CSE465) | In Progress

Python Qwen2.5-Omni-7B QLoRA Whisper Gradio EPITOME Framework

- Fine-tuning Qwen2.5-Omni-7B on EmpatheticDialogues using QLoRA (4-bit, rank 16) to produce measurably more empathetic responses in real-time voice conversations.
- Only the 7B language backbone is trained; Whisper audio encoder and speech decoder stay frozen, keeping training feasible on limited hardware.
- Empathy quality evaluated via EPITOME (LLM-graded 0-6), WER on IEMOCAP/RAVDESS, and a human Likert-scale study across 20 audio samples.

Zoco - Full-Stack E-Commerce Platform

Personal Project | Deployed on Vercel and Supabase

Next.js 14 TypeScript PostgreSQL Prisma ORM NextAuth.js Tailwind CSS

- Clothing e-commerce platform with a customer storefront and an admin dashboard covering inventory by size variant, order tracking, and sales analytics.
- NextAuth.js for auth, SSR with SEO metadata, Cloudinary for product images, and Zod for form validation.

Smart Patient Follow-Up and Health Trend Analytics

Junior Capstone (CSE299) | North South University

Node.js Express.js MySQL n8n Automation JWT Bootstrap

- Automates patient follow-up for local clinics: schedules appointments and sends Email and SMS reminders through n8n workflows.
- Classifies patients as Critical, Moderate, or Normal based on vitals, and generates downloadable PDF health reports for providers.

Intelligent Task Manager

Software Engineering Project (CSE327)

PHP MySQL JavaScript Bootstrap

- Rule-based scheduler that sorts academic tasks by deadline, duration, and priority. Ships with user auth, an admin panel, and a dashboard showing task completion progress.